	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	1132 - __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	(16320, 1, 1)x(256, 1, 1)	341.34 us	850,912	0 - NVIDIA GeForce RTX 5060	2.49 Ghz	[30684] trtexec.exe	GPU Compute Capability: 12.0 Registers/thread: 42 # SMs: 30 # TPCs: 15

GPU Speed Of Light Throughput

GPU Throughput Breakdown

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	53.92	Duration [us]	341.34
Memory Throughput [%]	65.59	Elapsed Cycles [cycle]	850,912
L1/TEX Cache Throughput [%]	51.64	SM Active Cycles [cycle]	830,550.27
L2 Cache Throughput [%]	45.80	SM Frequency [Ghz]	2.49
DRAM Throughput [%]	65.59	DRAM Frequency [Ghz]	13.79

High Memory Throughput Memory is more heavily utilized than Compute: Look at the Memory Workload Analysis section to identify the DRAM bottleneck. Check memory replay (coalescing) metrics to make sure you're efficiently utilizing the bytes transferred. Also consider whether it is possible to do more work per memory access (kernel fusion) or whether there are values you can (re)compute.

Key Performance Indicators

Metric Name	Value	Guidance
gpu__compute_memory_throughput.avg.pct_of_peak_sustained_elapsed	65.5937	65.594 - 53.920 >= 10.000

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 0% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

Compute Throughput Breakdown

SM: Pipe Fmaheavy Cycles Active [%]	53.92
SM: Inst Executed [%]	43.29
SM: Issue Active [%]	43.29
SM: Inst Executed Pipe Adu [%]	36.89
IDC: Request Cycles Active [%]	34.84
SM: Inst Executed Pipe Lsu [%]	33.81
SM: Pipe Aluheavy Cycles Active [%]	26.02
SM: Mio Inst Issued [%]	23.57
SM: Pipe Fma Cycles Active [%]	20.65
SM: Mio Pq Read Cycles Active [%]	8.20
SM: Inst Executed Pipe Xu [%]	8.20
SM: Mio Pq Write Cycles Active [%]	8.20
SM: Mio2rf Writeback Active [%]	6.54
SM: Memory Throughput Internal Activity [%]	1.02
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.51
SM: Inst Executed Pipe Ipa [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Inst Executed Pipe Uniform [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Pipe Tensor Cycles Active [%]	0
SM: Pipe Tma Cycles Active [%]	0

Memory Throughput Breakdown

DRAM: Cycles Active [%]	65.59
DRAM: Dram Sectors [%]	50.75
L1: Data Pipe Lsu Wavefronts [%]	50.50
L2: Xbar2lts Cycles Active [%]	45.80
L1: Lsuin Requests [%]	33.81
GPU: Compute Memory Access Throughput Internal Activity [%]	32.66
L2: T Tag Requests [%]	32.59
L1: Lsu Writeback Active [%]	23.34
L2: T Sectors [%]	21.25
L2: D Sectors [%]	18.87
L2: D Sectors Fill Device [%]	17.89
L1: M L1tex2xbar Req Cycles Active [%]	16.44
L2: Lts2xbar Cycles Active [%]	8.95
L1: M Xbar2l1tex Read Sectors [%]	6.40
L1: Data Bank Reads [%]	2.09
L1: Data Bank Writes [%]	1.63
SYSLTS: Xbar2syslts Cycles Active [%]	0.02
GPU: Compute Memory Request Throughput Internal Activity [%]	0.02
SYSLTS: T Sectors [%]	0.02
SYSLTS: T Tag Requests [%]	0.01
SYSLTS: D Sectors [%]	0.01
SYSLTS: Syslts2xbar Cycles Active [%]	0.01
SYSLTS: D Sectors Fill Sysmem [%]	0.01
L1: Texin Sm2tex Req Cycles Active [%]	0.00
L1: Data Pipe Tex Wavefronts [%]	0
L1: F Wavefronts [%]	0
L1: M Xbar2l1tex Read Sectors Mem Dshared [%]	0
L1: Tex Writeback Active [%]	0
L1: Tmain Requests [%]	0
L2: D Atomic Input Cycles Active [%]	0
L2: D Decomp Input Sectors [%]	0
SYSLTS: D Atomic Input Cycles Active [%]	0

Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.73	SM Busy [%]	53.92
Executed Ipc Active [inst/cycle]	1.77	Issue Slots Busy [%]	43.29
Issued Ipc Active [inst/cycle]	1.77		

Balanced Shared FMA Heavy is the highest-utilized pipeline (53.9%) based on elapsed cycles in the workload, taking into account the rates of its different instructions. It is a physical pipe and shared by the logical pipes FMA Heavy and ALU Lite. It's dominated by its FMA Heavy sub-pipeline. It is well-utilized, but should not be a bottleneck. Based on the number of executed instructions, the highest utilized pipeline (46.2%) is Shared FMA Heavy. It can execute instructions from FMA. Comparing the two, the overall pipeline utilization appears to be caused by frequent, low-latency instructions. See the Profiling Guide or hover over the pipeline name to understand the workloads handled by each pipeline. The Instruction Statistics section shows the mix of executed instructions for this workload.

Key Performance Indicators

Metric Name	Value	Guidance
sm__pipe_fmaheavy_cycles_active.avg.pct_of_peak_sustained_elapsed	53.9197	Try to decrease the utilization of the busiest pipeline (currently: Shared FMA Heavy)

Aggregate Pipe Utilization (% of elapsed cycles)

Aggregate Pipe Utilization (% of peak instructions executed over elapsed cycles)

Physical Pipe Utilization (% of elapsed cycles)

Physical Pipe Utilization (% of peak instructions executed over elapsed cycles)

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]	289.37	Mem Busy [%]	50.50
L1/TEX Hit Rate [%]	84.56	Max Bandwidth [%]	65.59
L2 Hit Rate [%]	41.46	Mem Pipes Busy [%]	36.89
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Shared Memory Spilling Requests	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Persisting Size [Mbyte]	4.72	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	0	-	-

Memory Chart

Values: Throughput
Inactivity: Greyed Out

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	16,320	Function Cache Configuration	CachePreferNone
Cluster Size	0	Preferred Cluster Size	0
Registers Per Thread [register/thread]	42	Cluster Scheduling Policy	PolicySpread
Static Shared Memory Per Block [byte/block]	0	Block Size	256
Dynamic Shared Memory Per Block [byte/block]	0	Threads [thread]	4,177,920
Driver Shared Memory Per Block [Kbyte/block]	1.02	Waves Per SM	108.80
Shared Memory Configuration Size [Kbyte]	16.38	Uses Green Context	0
Stack Size	1,024	# SMs [SM]	30
# TPCs	15	Enabled TPC IDs	all

WorkID/CLC Info

	Average	Min	Max	Sum
WorkId/CLC Requests	0	0	0	0
WorkId/CLC Requests Granted	0	0	0	0
WorkId/CLC Requests Granted as CTAs	0	0	0	0

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	83.33	Block Limit Registers [block]	5
Theoretical Active Warps per SM [warp]	40	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	79.88	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	38.34	Block Limit SM [block]	24
Cluster Occupancy [%]	0	Block Limit Barriers [block]	24
Max Active Clusters [cluster]	0	Max Cluster Size [block]	8
Overall GPU Occupancy [%]	0		

Impact of Varying Register Count Per Thread

Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block

Impact of Varying Block Barriers

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	830,550.27	Average L1 Active Cycles [cycle]	830,550.27
Average L2 Active Cycles [cycle]	744,885.50	Average SMSP Active Cycles [cycle]	831,539.53
Average DRAM Active Cycles [cycle]	3,086,704	Total SM Elapsed Cycles [cycle]	25,483,660
Total L1 Elapsed Cycles [cycle]	25,483,660	Total L2 Elapsed Cycles [cycle]	9,122,568
Total SMSP Elapsed Cycles [cycle]	101,934,640	Total DRAM Elapsed Cycles [cycle]	18,823,168

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	830,550.27	828,544	832,715	24,916,508
SMSP Active Cycles	831,539.53	828,123	833,638	99,784,744
L1 Active Cycles	830,550.27	828,544	832,715	24,916,508
L2 Active Cycles	744,885.50	744,699	745,165	8,938,626
DRAM Active Cycles	3,086,704	3,086,176	3,087,552	12,346,816

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	130,560	Branch Efficiency [%]	0
Branch Instructions Ratio [%]	0.00	Avg. Divergent Branches [branches]	0

Uncoalesced Global Accesses Est. Speedup: 59.93% This workload has uncoalesced global accesses resulting in a total of 8225280 excessive sectors (61% of the total 13447680 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

Metric Name	Value	Guidance
derived__memory_l2_theoretical_sectors_global_excessive	8.22528E+06	Reduce the number of excessive wavefronts in L2

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0xb010f5240 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	913,920	11
0xb010f5000 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	913,920	11
0xb010f47b0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	913,920	11
0xb010f46f0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	913,920	11
0xb010f4580 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f54ae97fff2219109002	913,920	11

Warp Stall Sampling (Not-issued Samples)

Location	Value	Value (%)
0xb010f5270 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	2,642	23
0xb010f5070 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	2,464	21
0xb010f4840 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	1,694	14
0xb010f55f0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	1,515	13
0xb010f4f60 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	591	5

Most Instructions Executed

Location	Value	Value (%)
0xb010f5610 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	130,560	0
0xb010f5600 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	130,560	0
0xb010f55f0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	130,560	0
0xb010f55e0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	130,560	0
0xb010f55d0 in __myl_MulMinMaxRounCastConc_0xfb670565fd23f...	130,560	0

Still missing data? Collect the *full* metric set, enable individual sections or learn how to select individual metrics for collection.